

Retail Data Warehouse & Power BI Analytics

End-to-End Business Intelligence Project

SQL • ETL • Data Warehouse • Power BI • DAX

Prepared By

Gehad Mosaed

Project Overview

The project covers:

- **Data Warehousing Design**
- **ETL Pipeline Development**
- **Business Query Analysis**
- **Power BI Dashboard Development**
- **KPI Tracking and Business Insights**

Project Objectives

This project demonstrates the complete development of a Retail Data Warehouse and Business Intelligence solution.

Technologies Used:

MySQL 

ETL



Power BI 

Star Schema

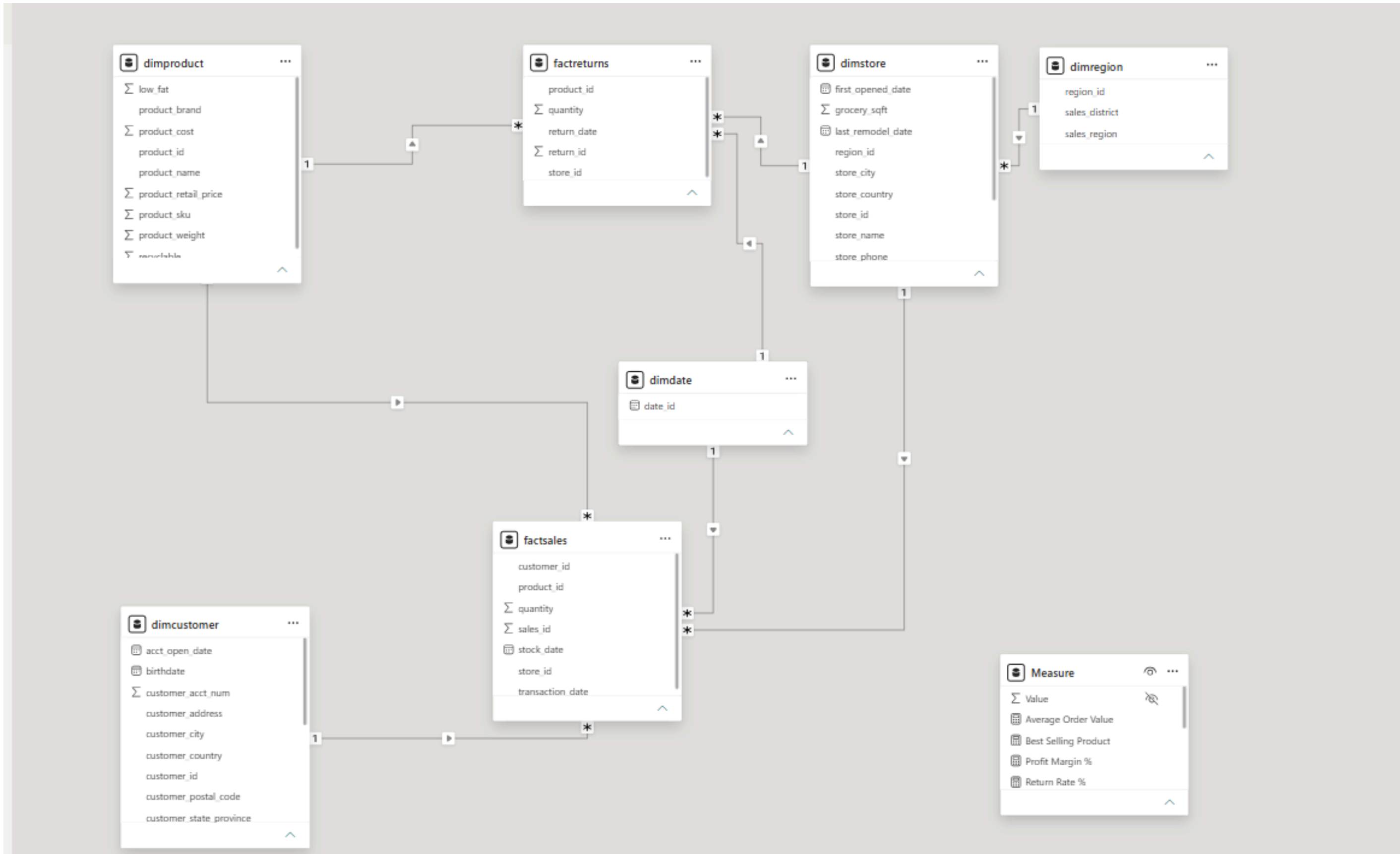


DAX 

Data Warehouse



Data Warehousing Design



ETL Pipeline Development

The project was built using a structured ETL workflow:

Extract



Transform



Load

Data imported from
source retail datasets.

Data cleaning
Data validation
Standardization
Key generation

Data loaded into
dimensional warehouse
tables.

Business Value

Business Impact

This dashboard enables stakeholders to:

- Monitor sales performance.
- Track profitability.
- Identify high-performing products.
- Understand customer behavior.
- Monitor returns and operational efficiency.
- Support data-driven decision making.

Retail

Executive Overview

Product Analysis

Customer Analysis

Returns & Profitability

Executive Overview

Total Sales

 1.76M

Total Orders

 270K

Return Rate %

 0.99%

Total Profit

 1.05M

Sales Region

All

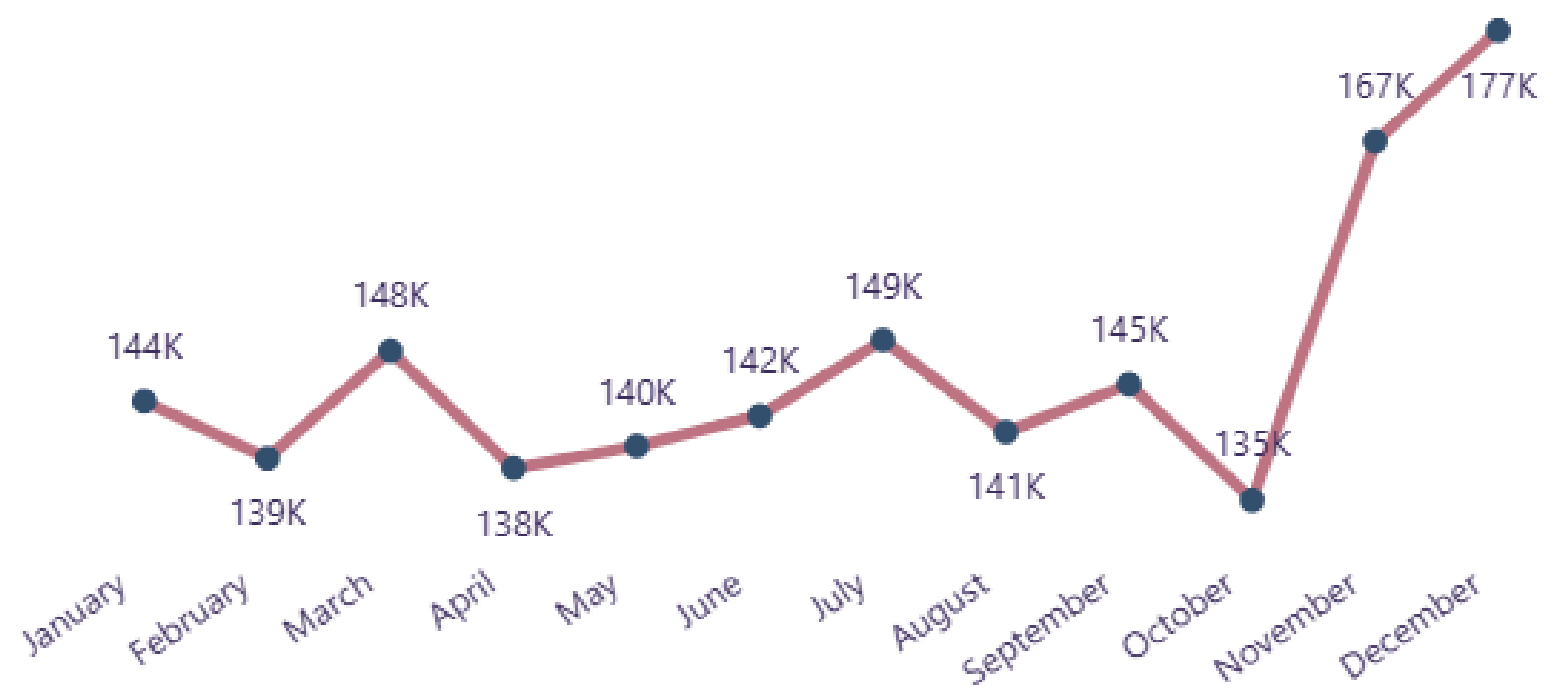
Year

All

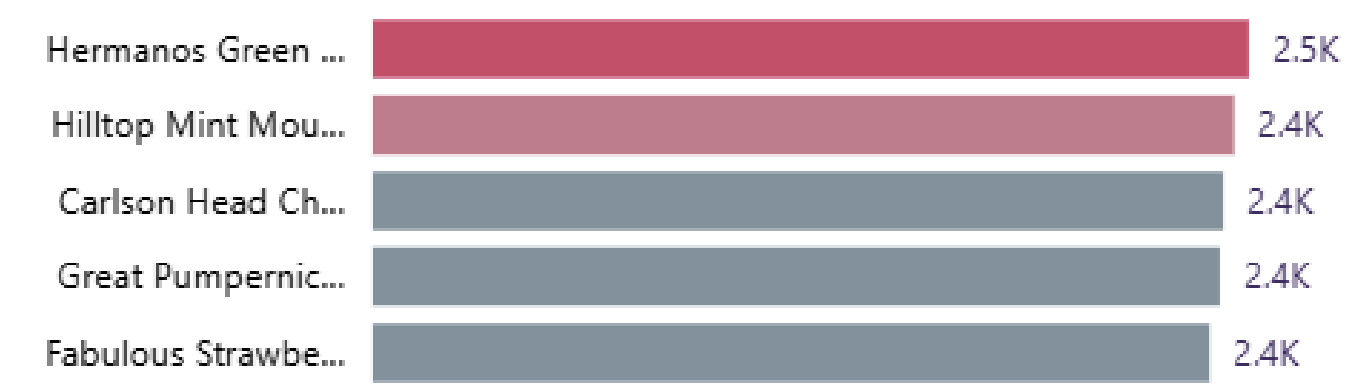
Store Type

All

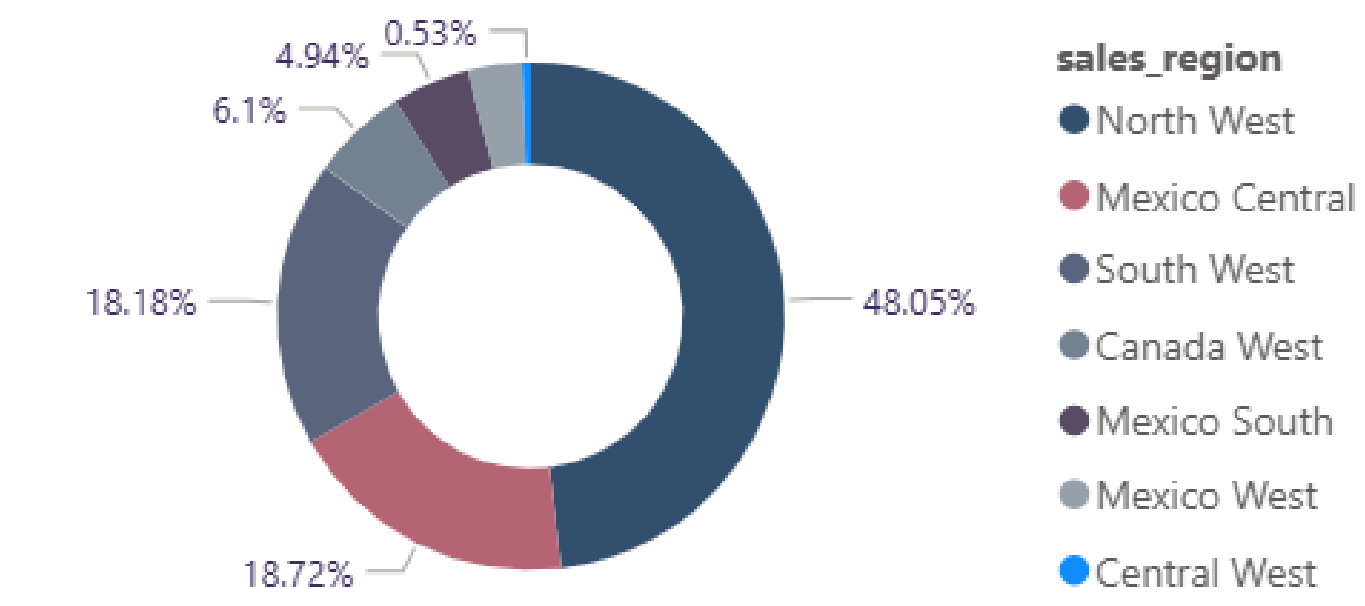
Sales Trend Over Time



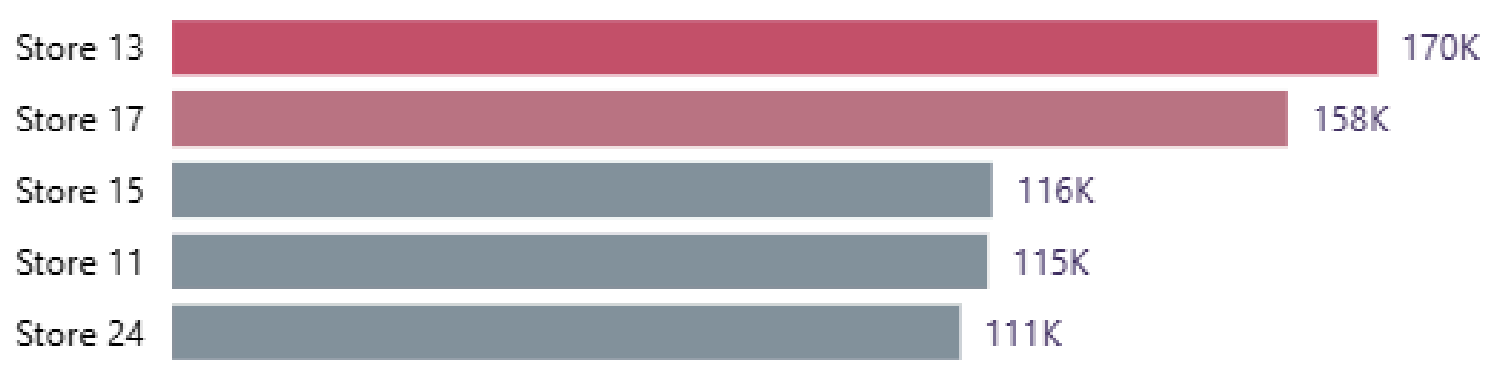
Top 5 Products



Sales by Region



Top 5 Stores



Product Analysis

Retail

Executive
Overview

Product
Analysis

Customer
Analysis

Returns &
Profitability

Total Sales



1.76M

Total Profit



1.05M

Total Products



1.56K

Best Selling Product



Hermanos Green Pepper

Product Performance

product_name	Total Sales	Total Profit	Total Units Sold	Profit Margin %
Hermanos Green Pepper	2,489.70	1,670.55	645	67%
Imagine Popsicles	2,160.27	1,513.89	567	70%
Hilltop Mint Mouthwash	2,447.12	1,493.96	676	61%
Super Extra Chunky Peanut Butter	2,211.82	1,484.73	611	67%
Nationeel Potato Chips	2,112.00	1,479.50	550	70%
Fabulous Strawberry Drink	2,377.90	1,474.90	602	62%
Booker Strawberry Yogurt	2,133.28	1,452.56	536	68%
BBB Best Apple Preserves	2,184.05	1,439.90	605	66%
Hermanos New Potatos	2,103.58	1,431.08	538	68%
High Quality Scented Toilet Tissue	2,079.04	1,413.28	584	68%

Top 10 Products by Profit

Hermanos Green Pepper	1671
Imagine Popsicles	1514
Hilltop Mint Mouthwash	1494
Super Extra Chunky Pe...	1485
Nationeel Potato Chips	1480

Top 10 Products by Sales

Hermanos Green ...	2.5K
Hilltop Mint Mout...	2.4K
Carlson Head Che...	2.4K
Great Pumpernick...	2.4K
Fabulous Strawbe...	2.4K
Fort West BBQ Po...	2.3K

Top 10 Product Brands

product_brand	Total Sales	Total Profit	Total Units Sold	Profit Margin %
Hermanos	56,659.48	33,167.27	24682	59%
Tell Tale	51,561.22	29,925.59	23801	58%
Ebony	49,727.46	29,749.45	23492	60%
Tri-State	49,347.75	29,064.76	23400	59%
High Top	47,200.53	28,502.83	22040	60%
Nationeel	45,384.82	27,446.16	19959	60%
Horatio	43,787.75	25,589.28	19205	58%
Fast	40,532.84	24,746.78	19118	61%
Best Choice	42,738.02	25,901.32	18553	61%

Retail

Executive
Overview

Product
Analysis

Customer
Analysis

Returns &
Profitability

Customer Analysis

Total Sales

1.76M

Total Orders

270K

Total Customers

8.84K

Avg Order Value

\$6.54

Year

All

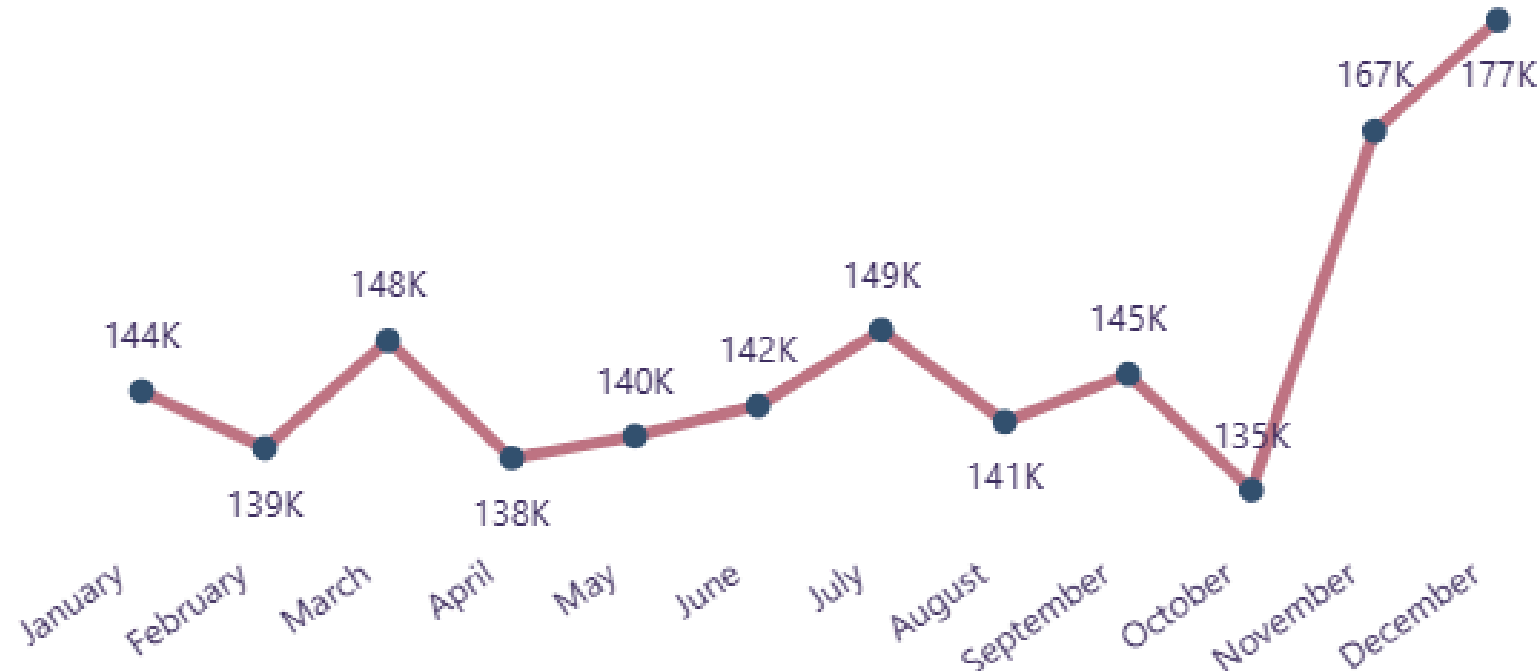
Store Type

All

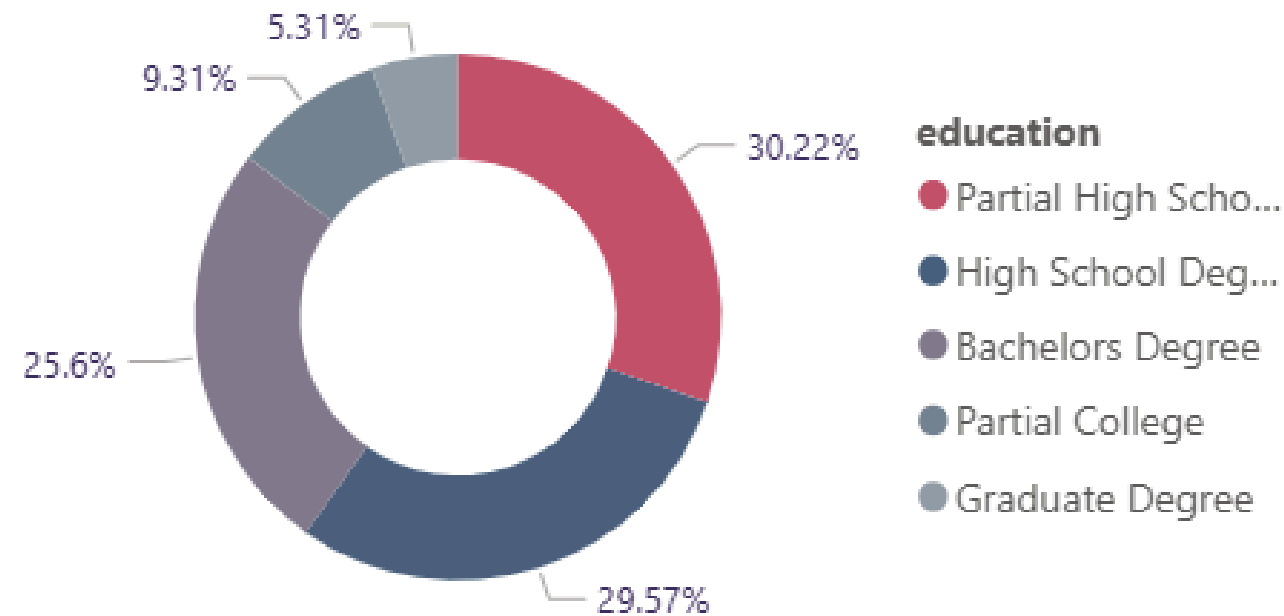
Sales Region

All

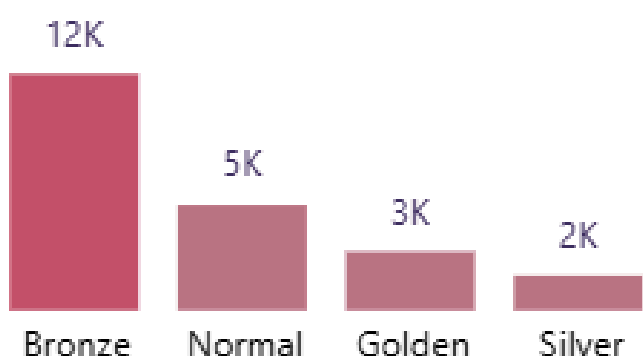
Sales Trend Over Time



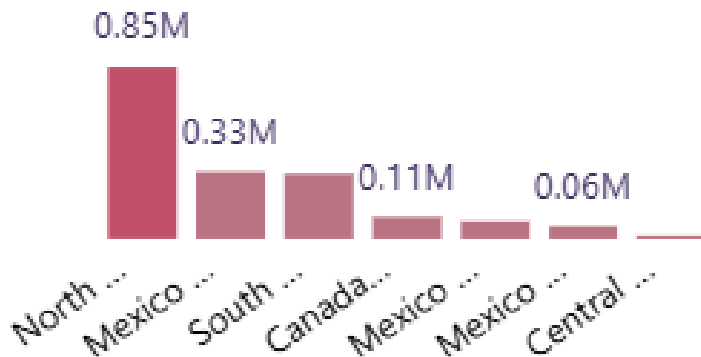
Sales by Education



Customer Segmentation



Sales by Region



Top Customers

first_name	last_name	customer_city	Total Sales	Total Orders
Dawn	Laner	Spokane	1,995.21	301
Ida	Rodriguez	Spokane	2,235.43	290
James	Horvat	Spokane	2,121.31	297
Mary Francis	Benigar	Spokane	1,973.79	275
Wildon	Cameron	Spokane	1,955.25	233

Retail

Executive
Overview

Product
Analysis

Customer
Analysis

Returns &
Profitability

Returns & Profitability

Total Profit

1.05M

Profit Margin %

60%

Return Rate %

0.99%

Total Returns

8K

Year

All

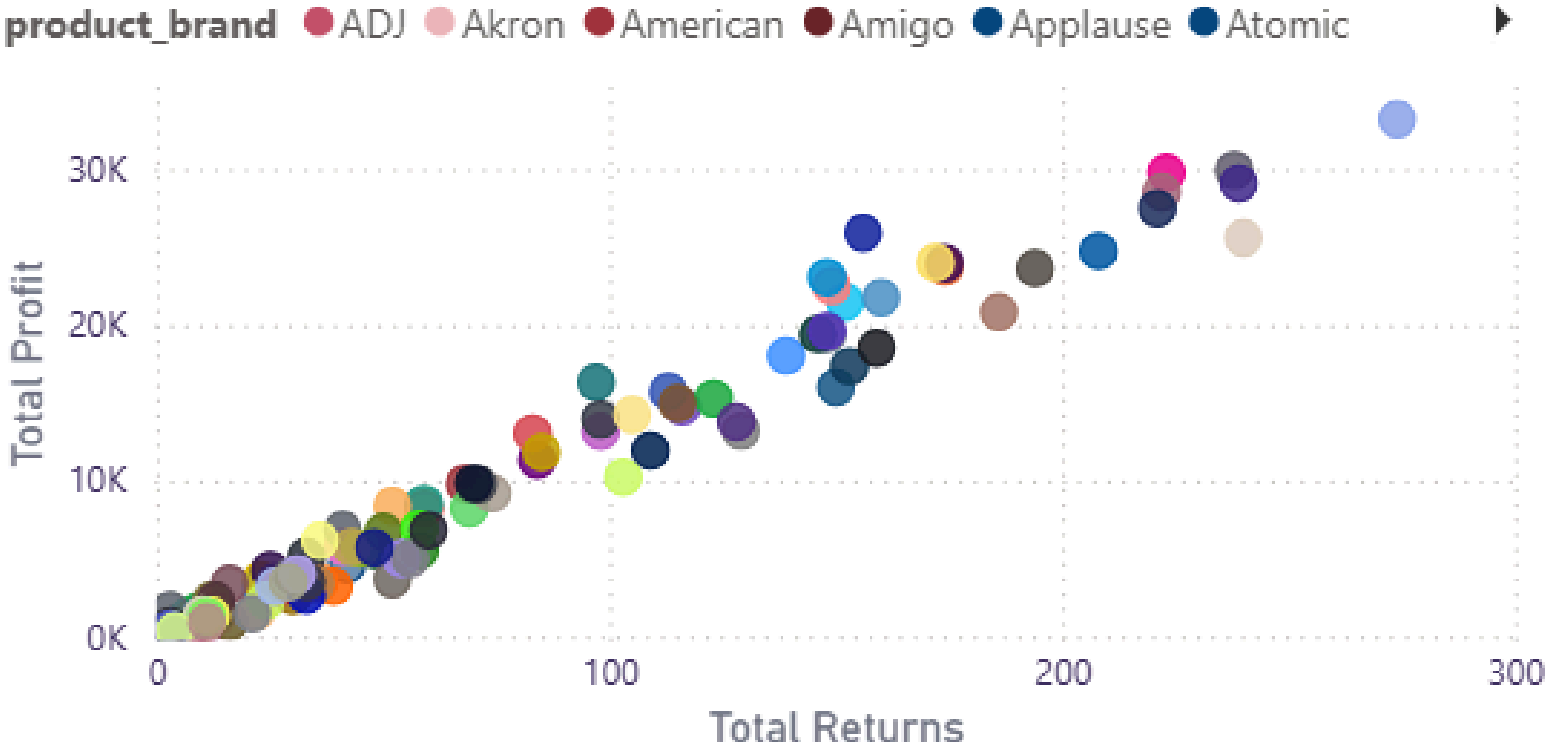
Product Brand

All

Sales Region

All

Market Performance by Country



Conclusion

01

Project Summary

This project demonstrates the end-to-end development of a modern Retail Analytics solution, beginning with Data Warehousing and ETL development and ending with interactive Power BI dashboards that deliver actionable business insights.

02

Skills Demonstrated

- SQL Development
- Data Warehousing
- ETL Design
- Star Schema Modeling
- DAX Calculations
- Power BI Dashboard Design
- Business Intelligence Reporting